Drift to Default: How Autoregressive Video Rollouts Forget Their Opening

Anonymous authors

Abstract

Autoregressive video diffusion models generate long videos chunk by chunk and drift over tens of seconds. We measure drift by intervention: we shift the brightness or colour of a held opening frame within a scene, or start from openings of different scene classes, and measure how much of the opening survives. In 25 preregistered studies on nine open models (1.3B to 14B parameters, three model families), rollouts *relax*: global tone and scene content move from the opening toward a default set by the conditioning, keeping a share α that depends on the model, the prompt and the horizon. Under a generic city prompt, openings of six scene classes lose 39–93% of their class evidence within 30 s. How much is kept depends on whether the opening stays in the model's context. Once it leaves, it is forgotten, through a sliding key–value window at 1.3B and at 14B parameters, and after a delay of about 20 s when overlapping frames are re-denoised. An opening that stays in the context keeps a share whose size depends on the model: 0.2–0.5 of its tone at 60 s for multi-frame sinks, anchors and clean prefixes [study 24: at 180 s], while a single anchored latent frame keeps almost no tone (0.03–0.08) but, in the 14B model, a third of the scene. Attention sinks left at their absolute rotary positions keep more, much of it in bursts [study 25: at query–sink distances predicted by an attention probe, which move when the sink is moved]. In an exploratory analysis, drift separates into forgetting the opening and an offset of the default from where the same prompt starts, small in the two models trained on their own long rollouts. A content-aware correction and a deeper sink reduce drift without training. We report all N preregistered primary tests, M of which failed.

1 Introduction

Autoregressive video diffusion models generate a video chunk by chunk, each chunk conditioned on the model's own earlier output [3, 1, 2]. This makes streaming and open-ended generation possible and is the basis of interactive world models [38, 39, 40]. It also makes them drift: over tens of seconds the picture darkens or saturates, loses texture, and its content wanders. A growing family of methods targets this failure with rolling windows, attention sinks, re-aligned positions, longer self-rollouts and test-time correction [7, 8, 9, 22, 12, 13, 21]. Most report a few samples per prompt and an aggregate quality score. What drift *is*, in quantitative terms, is rarely measured.

We measure it by intervention. We hold an opening frame, shift its brightness or colour *within* a scene, let the model continue, and ask how much of the shift survives; for scene content we start from openings of different classes. In every model we tested the rollout *relaxes*: it moves from its opening toward a default state set by the conditioning and keeps a share α of where it started that depends on the model, the prompt and the horizon,

$$x(T) \approx \alpha(T)\, x_{\text{open}} + \big(1 - \alpha(T)\big)\, \mu. \tag{1}$$

This is a linear-response summary, not a law: α and μ come from one regression. What it organises are three findings, and tests of its assumptions (Section 4).

What survives depends on whether the opening stays in the context, not on model size (Fig. 3). Once the opening leaves a sliding key–value window it is forgotten, in 1.3B models and in a 14B model alike; re-denoising overlapping frames passes it forward and delays forgetting by about 20 s. An opening that stays in the context (a re-rotated attention sink, an anchor, a clean copy in every section, full history) keeps a share whose size depends on the model [and that levels off / keeps falling to 180 s], and tone and scene can be kept separately: a one-frame anchor in the 14B model keeps the scene but not its tone, LongLive's

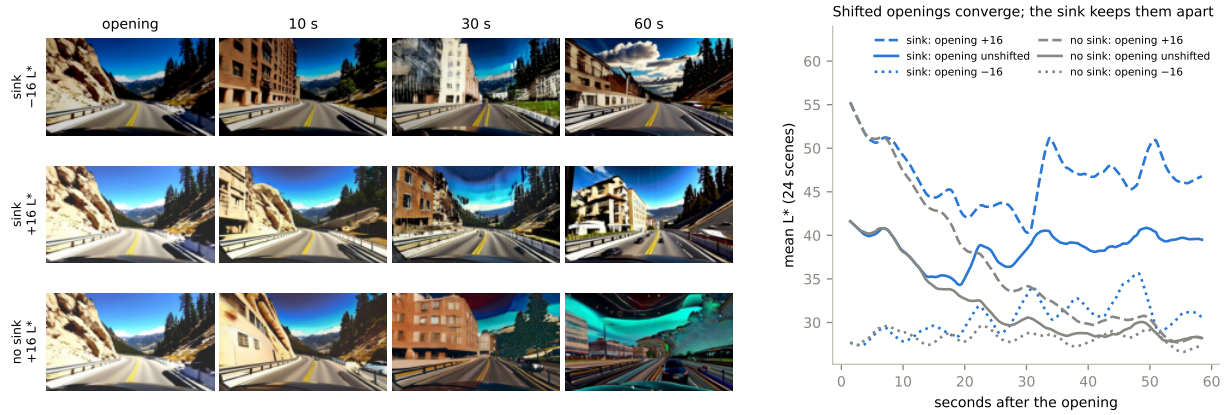


Figure 1: Relaxation. Left: a generated mountain-road opening, darkened or brightened by 16 L* before encoding, continued for 60 s under a generic city prompt by Self Forcing DMD with and without an attention sink. The scene turns into a city street in all three runs; without a sink the colours also blow out. Right: mean brightness over the 24 scenes of study 7 for the three versions of each opening (3-s moving average). Without a sink the versions converge to one default; with a sink they stay apart by a share that fluctuates around half.

sink the reverse. A sink left at its absolute rotary positions keeps more, much of it in periodic bursts in which the opening, including its scene, returns [at distances an attention probe predicts].

For a given checkpoint, the default is set by the conditioning. Lighting words move the default brightness by up to 27 L*, and scene words decide whether a snowy, desert or forest opening stays or becomes a city street within 30 s. Which parts of the conditioning carry tone and which carry the scene differs between models.

Drift has two separable parts (exploratory): forgetting the opening, $(1-\alpha)(x_{\text{open}}-\mu)$, and an offset of the default itself from where the same prompt starts, which is small in the two models trained on their own long rollouts and 4–20 L* in the others we measured. Two training-free levers act on these terms: an early-step tone correction and a deeper sink, each with its own cost; lighting words matched to a model's measured defaults helped on one model only.

Contributions.

1. A measurement protocol for long-horizon drift: held and perturbed openings, persistence α and default μ for tone and scene content, paired seeds pinned to one GPU architecture, and preregistered analyses.
2. An empirical account of relaxation across nine open models (six of the Wan2.1-1.3B architecture, the Wan2.1-14B student Krea Realtime, the HunyuanVideo-based FramePack and MAGI-1), for tone and scene content, at horizons up to 180 s: what sets μ, what sets the long-run share, and an exploratory decomposition of drift into forgetting and attractor offset.
3. A positional account of attention-sink bursts [study 25].
4. Training-free levers matched to the two terms, compared on the same prompts and seeds, with their costs and failures.

We preregistered 25 studies with hashed analysis code and report all N primary tests, including the M that failed (Table 3). No model is trained; all code and per-video statistics are public.

2 Related work

Autoregressive and long video diffusion. Diffusion Forcing [3] and History-Guided Video Diffusion [33] enable causal rollout with per-frame noise and guidance on the history. CausVid [1] distils a bidirectional teacher [4] into a few-step causal student with DMD [5]; Self Forcing [2] and Self-Forcing++ [22] train on the student's own (minute-long) rollouts. LongLive [7], Rolling Forcing [8] and Deep Forcing [9] add attention

sinks, rolling joint denoising and deeper, re-aligned sinks; LoL [21] traces "sink collapse" to positional periodicity. Segment-wise models condition on overlapping frames [10, 26, 31], a temporal pyramid [30] or packed history [23]; training-free schemes lengthen bidirectional models [25, 27, 28, 29]. These works propose mechanisms against drift; we measure what drift does and which mechanism changes which part of it.

Error accumulation, sinks and correction. Sampling from one's own output causes exposure bias in sequence models [34, 35] and diffusion models [36, 37]; error analyses of autoregressive video diffusion [14] bound how errors compound, and world models [38, 39, 40, 50] add noise to context frames for robustness. For global statistics our results suggest a complementary view: the rollout forgets its context and relaxes to the conditioning's default. Global brightness biases of diffusion models arise at the earliest denoising steps [41, 42]. Streaming language models keep the first tokens as an attention sink [11, 43], related to registers in vision transformers [44]; the closest analogue to our finding is instruction drift in dialogs [45], where attention to the system prompt decays and behaviour reverts to a default. Pathwise test-time correction [12] and FreqForcing [13] anchor rollouts to early frames or their low frequencies; our correction is a minimal member of this family. VBench [15] scores imaging quality with MUSIQ [16]; we measure drift against each prompt's fresh-start distribution [17] and by intervention.

3 Measuring drift by intervention

Models. We study nine open models (Appendix Table 2). Self Forcing (five checkpoints), CausVid, LongLive and Rolling Forcing are causal few-step students of Wan2.1-T2V-1.3B [4] that read their past through a key–value cache of 21 latent frames (LongLive: 12), optionally with an attention sink (the first chunk kept in the cache); Krea Realtime 14B [32] is a student of the 14B model. SkyReels-V2-DF [10] (same architecture as the 1.3B model) extends videos in overlapping segments, Matrix-Game-2.0 [50] is a Wan-architecture world model driven by actions, FramePack [23] (13B, from HunyuanVideo [24]) conditions every section on the opening, and MAGI-1 [31] (4.5B) has its own VAE and transformer; Section 5 describes how each holds the opening. All Wan-1.3B cache configurations run in one harness that reproduces the official Self Forcing loop bit for bit when no intervention is applied.

Held and perturbed openings. An opening image is held still for the first chunk (9 frames, 0.56 s at 16 fps; SkyReels: 17 frames as a prefix video; FramePack and Krea: the input image; MAGI-1: a clean image prefix) and continued under a prompt. Openings are NAVSIM [19] front-camera frames or frames generated by the models themselves. To separate what the model keeps from what the scene is, we shift the *same* opening in CIELAB before encoding: L^* by ± 8 and ± 16, b^* by ± 12. For scene content, we instead generate openings of six scene classes and vary the prompt (Section 4.1).

Persistence and default. For a statistic x (a per-frame CIELAB mean), the persistence at time T is the within-scene least-squares slope of the end value $\bar{x}(T)$, averaged over the window $[T-w, T)$ after the opening, on the opening value x_{open} (mean of the held frames),

$$\alpha(T) = \frac{\partial\, \bar{x}(T)}{\partial\, x_{\text{open}}}\Big|_{\text{scene}}, \qquad \mu(T) = \frac{b_0}{1-\alpha(T)}, \tag{2}$$

where b_0 is the fitted intercept, so μ is the level at which the end equals the opening (the zero-drift level). The window is 10 s for endpoints at 30 s or later (2–5 s in the earlier studies with shorter endpoints) and 1 s for curves; Appendix Table 4 compares all models over one window, 15–20 s. Averaging slopes over scenes gives a scene-bootstrap interval. The fresh default is where prompt-only rollouts end.

Content persistence. For scene content we score each frame with CLIP ViT-L/14 [48] as the log-odds of the opening's class against "a city street", $z = s\,(\cos(f, t_{\text{class}}) - \cos(f, t_{\text{city}}))$, and define $\kappa(T) = (z(T) - \mu_z)/(z_{\text{open}} - \mu_z)$ with μ_z the class score of fresh rollouts of the generic prompt over the same window. κ is the content analogue of α: 1 if the class evidence is kept, 0 if the rollout looks like a fresh generic rollout.

Drift metrics for prompt-only rollouts. Where no opening is held, drift is measured against fresh 5-s generations of the same prompt, in colour statistics (D) and DINOv2 [17] features (K; Appendix B).

Pairing and preregistration. With bf16 kernels, the same seed and code give bit-identical videos within a GPU architecture but different videos across architectures, so arms that are compared run in one job, which regenerates its outputs if a preempted job resumes on another architecture. Each study's hypotheses, decision rules and analysis code were frozen by SHA-256 hash before its videos were generated (study 1: before analysis), and studies 3–25 were committed publicly first. Tests within a study are Holm-corrected; Table 3 lists every primary test.

4 Rollouts relax toward a conditioning-set default

Tone. Observationally, the 31-s change of every global statistic is anti-correlated with its opening value (r from -0.79 to -0.95; Appendix Table 5), but openings differ in content too, so we intervened: we shifted the brightness and colour temperature of the same opening within each of 24 scenes (12 NAVSIM, 12 generated) and continued every version with the generic prompt (study 6, 31 s; study 10, 60 s). The versions converge (Fig. 1). At 60 s every cache configuration has relaxed (Appendix Fig. 5a): α ranges from -0.01 (LongLive without a sink) to 0.79 (Self Forcing GAN), ranked as at 31 s (Spearman $\rho = 0.78$, $p = 0.008$). Part of the loss is immediate: most configurations lose a fifth to a third of the shift within 4.5 s, and SiD loses two thirds there and nothing afterwards (0.66 by 4.5 s, 0.65 by 60 s), which looks like rejecting an off-manifold shift rather than forgetting it. Colour temperature behaves like brightness. Persistence varies by checkpoint even with architecture, context and sampler fixed (Self Forcing: 0.35 to 0.79 at 60 s; two SiD checkpoints differ by 0.36), so the training objective alone does not set it.

Is the response linear in the shift? Equation 1 treats α as independent of how far the opening was shifted, but every α above comes from one pair of shifts (± 16 L*). Study 23 shifted the same 24 openings by -24, -8, 0, $+8$ and $+24$ L* (Self Forcing, 60 s, two seeds, 720 videos) and tested equivalence within ± 0.15. [results: large vs small shift with sink, re-rotated sink; darkening vs brightening; no sink forgets a large shift too.] In study 6, persistence from ± 8 and from ± 16 shifts agrees within 0.09 in 12 of 14 configurations at 31 s (exploratory; Appendix C).

4.1 Scene content

Global tone is the easiest statistic to shift and measure, but it is not what a user sees as "the scene". We therefore generated openings of six scene classes (snow, desert, forest, coast, mountain, farmland; four each, selected by CLIP before any continuation) and continued each under the generic city prompt and under its own class prompt, with eight configurations (study 9). Content relaxes like tone (Fig. 2). Under the city prompt, κ at 30 s ranges from 0.07 (SiD) to 0.61 (Rolling Forcing), and the opening's class is CLIP's top label in only 34% of Self Forcing frames at 30 s. Under the opening's own class prompt the class persists (κ 0.71–1.05; 9.3 [8.1, 10.5] more log-odds than under the city prompt, preregistered), as a fresh rollout of that prompt would also show it; the informative result is the loss under the city prompt. Content and tone persistence tend to go together across configurations ($\rho = 0.67$, $n = 8$, $p = 0.07$). Class scores could partly reflect colour (snow is bright, sand is yellow), but scoring frames whose CIELAB mean and spread are matched to the opening's gives the same results, as does SigLIP (Appendix C).

Sinks act differently on content. Self Forcing's sink keeps more of it at 60 s (κ 0.61 vs 0.14, +0.48 [0.39, 0.57], preregistered), but not steadily: κ falls to 0.32 at 30 s and rises again, and the dunes, the sea and the mountains of the opening reappear after a minute of city (Fig. 2, left). The returns are positional (study 16, 24 openings): with the sink's keys re-rotated to sit just before the window, they vanish (burst contrast -0.29 [-0.37, -0.22]), and the content kept at 60 s falls from 0.56 to 0.25 (-0.31 [-0.37, -0.25]). With this one seed the re-rotated sink kept no detectably more content than no sink (+0.10 [-0.00, 0.21], not supported). Three new seeds per opening (study 21, 240 videos) reproduce the sink's advantage, the loss under re-rotation (-0.40 [-0.47, -0.34]) and the positional return, and, averaged per opening, detect a small share kept by the re-rotated sink: 0.19 against 0.11 without a sink (+0.08, Holm $p = 0.006$; [-0.01, 0.16] when the fresh starts

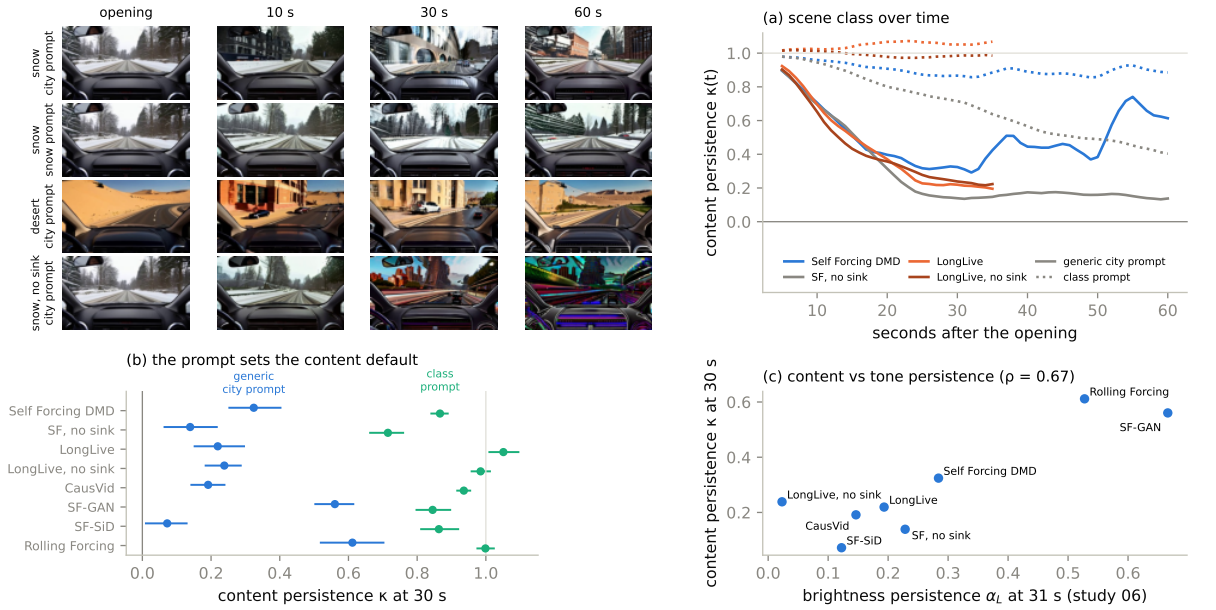


Figure 2: Scene content relaxes to the prompt's default (study 9). Left: a snow and a desert opening continued by Self Forcing DMD under the city prompt or their own class prompt. Under the city prompt both become city streets; with a sink the desert returns at 60 s; without a sink the video collapses. (a) Content persistence $\kappa(t)$ (CLIP log-odds of the opening's class against "city street", 0 = fresh generic rollout, 1 = opening) under the city prompt (solid) and the class prompt (dotted). (b) κ at 30 s, 24 openings, 95% CIs. (c) Content against brightness persistence across configurations.

are also resampled). What a Self Forcing sink keeps of the scene, it keeps mostly by periodically re-injecting the opening. LongLive's sink, which keeps tone, does not keep content (0.22 vs 0.24, preregistered test not supported).

Content relaxes outside the Wan-1.3B family too. FramePack, continuing the same 24 openings under the city prompt, keeps 0.55 of the class evidence at 20 s (Self Forcing: 0.40 with a sink, 0.31 without; study 17). Giving its two routes different images separates them (Appendix Fig. 7b): the clean context prefix alone keeps 0.24 $[0.12, 0.37]$, levelling off after 14 s, while the SigLIP image embedding alone, with a city frame in the context, *pulls* the video toward the opening's class, to 0.23 $[0.16, 0.30]$ at 20 s and still rising. The same embedding has no effect on tone (study 11), so in FramePack it sets the default of the scene but not of its brightness. Tone and content can also part the other way: Krea Realtime 14B's first-frame anchor keeps the opening's class (κ 0.37 vs 0.05 without it at 60 s, study 22; SigLIP and tone-normalised scores agree) but none of its tone (Section 5). [MAGI-1 (study 19) content results.]

4.2 The default

The conditioning sets it. Continuing the same 40 openings under four lighting prompts gives four defaults: on Self Forcing, sunny 43, overcast 34, dusk 19 and night 16 L*; on LongLive the same words give 45, 40, 32 and 21 (Appendix Fig. 8). The mapping from words to brightness is model-specific. Under one prompt the default also differs between checkpoints of one architecture (zero-drift levels from 17 to 54 L*), and it tracks where prompt-only rollouts end ($r = 0.93$, slope 1.58; Appendix C).

Persistence depends on the scene. Equation 1 separates a model term (α) from a prompt term (μ), but α itself varies: Self Forcing keeps 0.34 of the opening under a sunny prompt and 0.05 under a night prompt (exploratory interaction, $p = 0.001$). In four new scene types with their own prompts and self-generated openings (a park walk, a drone flight over a coast, a kitchen, a market; study 12), its persistence at 60 s differs between them (0.44 to 0.67, $p = 0.049$, preregistered), as does LongLive's at 30 s (0.10 to 0.43). The sink effect replicates almost exactly outside driving scenes: 0.55 with a sink and 0.02 without (+0.53 $[0.45, 0.61]$).

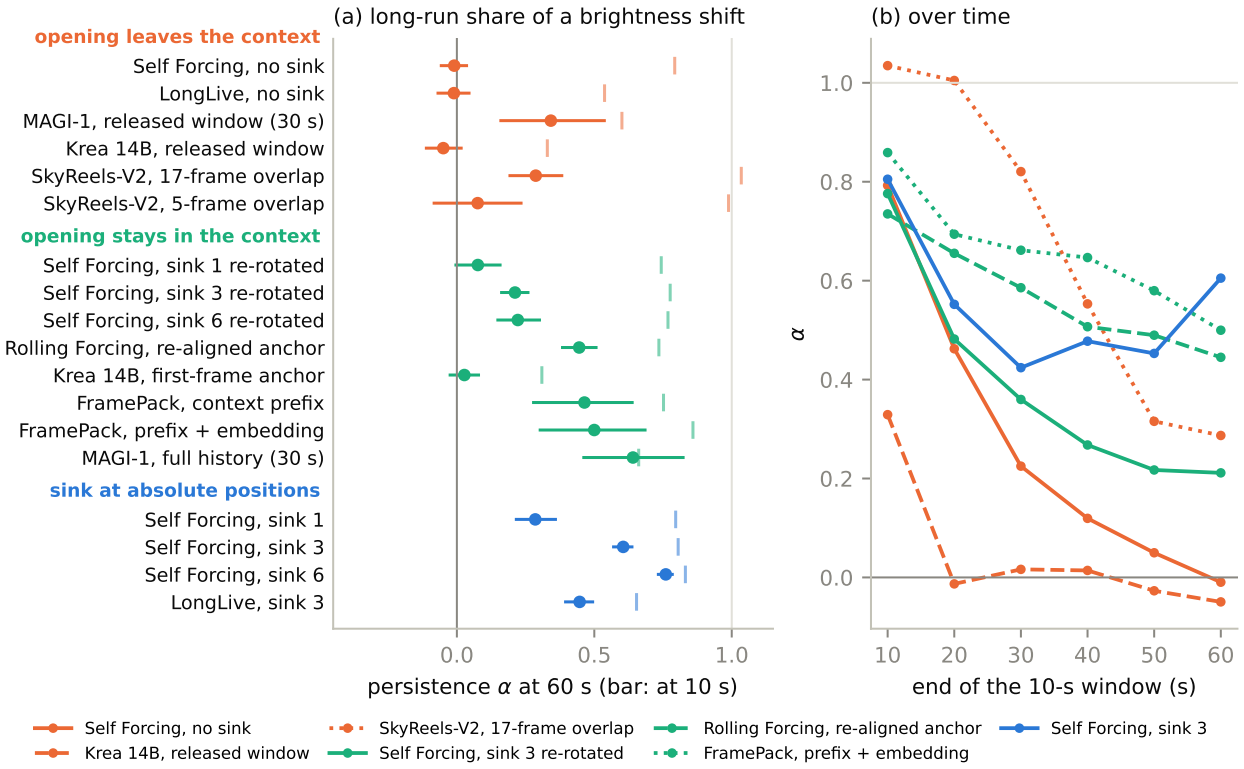


Figure 3: Whether the opening stays in the context decides how much of it is kept. (a) Persistence of an L* shift at 60 s (window [50, 60) s; MAGI-1 full history: 30 s) with scene-bootstrap 95% CIs; ticks: at 10 s. (b) Persistence over time (10-s windows) for one or two configurations per group.

Two components of drift. For a prompt-only rollout, Eq. 1 says nothing about why the default should differ from where the same prompt starts. It does, and by how much depends on training (Appendix Fig. 6a). Fresh rollouts of Self Forcing (trained on its own 5-s rollouts; a checkpoint trained on 10-s rollouts alike) and CausVid end 4–11 L* from where they start; those of LongLive and Rolling Forcing, trained or tuned on their own long rollouts, 0.7–2.3 L*. Yet LongLive still forgets its opening ($\alpha = 0.02$ without a sink at 31 s). In this exploratory analysis drift has two separable parts: *forgetting* the opening, $(1-\alpha)(x_{\text{open}} - \mu)$, which every attention-based model does, and an *attractor offset* $\mu - x_{\text{start}}$, which is near zero in the two models trained on their own long rollouts (other differences between them and the short-clip models cannot be excluded). Two later models fit this pattern: fresh rollouts of Krea 14B, distilled with Self Forcing, brighten by 4.8 L* over 31 s (8 seeds), and those of MAGI-1, trained chunk by chunk on real videos rather than on its own rollouts, by 15–20 L* over 30 s. Prompt-only drift metrics mix the two, and both keep rising over 31 s even with a sink (Appendix H); relaxation describes the memory of the opening, not the absence of degradation.

5 What decides how much of the opening is kept

Figure 3 collects every configuration we ran for 60 s, grouped by whether the opening stays in the model's context. Once it leaves, it is forgotten; while it stays, a share is kept whose size depends on the model; and sinks left at their absolute rotary positions add bursts in which the opening returns.

Once the opening leaves the context, it is forgotten. Without a sink, Self Forcing and LongLive keep 0.05 and −0.01 at 60 s (study 10). MAGI-1 [31] (4.5B, distilled), which shares neither architecture nor training with the other models, attends in its released configuration to the four previous 1-s chunks, so the opening leaves its context after about 4 s; persistence falls from 0.87 at 5 s to 0.34 at 30 s (−0.52 [-0.70, -0.35], study 18) and to [x] at 60 s (study 19). Krea Realtime 14B [32], a Self Forcing student of Wan2.1-14B,

recomputes its cache over a window of three latent frames (study 22, 24 scenes): it keeps 0.33 at 10 s and nothing from 20 s on (-0.05 at 60 s; -0.38 [-0.45, -0.31] from 10 to 60 s), and its class evidence falls to $\kappa = 0.05$. The pattern holds at 14B parameters, although Krea's window also differs, so this is not a controlled test of scale. Re-denoising overlapping frames delays forgetting but does not stop it. SkyReels-V2-DF extends videos in 97-frame segments that re-denoise the last 17 frames of the previous segment jointly with the new ones. It holds the opening for 20 s ($\alpha = 0.96$ [0.86, 1.07], whatever its guidance, history noise or number of steps, exploratory; with a 5-frame overlap it relaxes, -0.24 [-0.40, -0.09] at 15–20 s, preregistered) and then forgets steadily, to 0.29 at 60 s (-0.72 [-0.81, -0.63] from 20 to 60 s; 5 frames: 0.08; study 16).

An opening that stays in the context keeps a share. Self Forcing keeps the sink's keys at their absolute rotary positions, ever further from the current chunk. Re-rotated to sit just before the window, as in Rolling Forcing and Deep Forcing, the sink stays at query distances seen in training. Re-rotated sinks of three and six frames keep the same share at 60 s (0.23 and 0.22, -0.00 [-0.11, 0.10], study 16; the preregistered depth test is not supported) and one frame keeps 0.08 (study 10); with three new seeds per scene a three-frame sink keeps 0.21 against -0.01 without one ($+0.22$ [0.15, 0.29], study 20, in each seed alone too). Rolling Forcing, which re-aligns its anchor and was trained on longer rollouts, keeps 0.45. Krea's optional first-frame anchor, one latent frame next to its window, keeps scene content but not tone: $\kappa = 0.37$ against 0.05 without it ($+0.33$ [0.18, 0.47], preregistered; the opening's class is CLIP's top label in 37% of frames against 14%), but $\alpha = 0.03$ [-0.03, 0.08] (both tone tests not supported), close to the 0.08 of a one-frame re-rotated sink in Self Forcing. FramePack [23] (13B, fine-tuned from HunyuanVideo [24]) places the opening as a clean prefix frame in the context of *every* 33-frame section and also feeds its SigLIP embedding as a global condition. The shift travels through the prefix: sent only through the prefix it keeps about half at 20 s, only through the embedding nothing (0.01 [-0.14, 0.17], study 11); at 60 s FramePack keeps 0.51 [0.29, 0.71], and 0.46 [0.28, 0.65] with the prefix alone (study 16). MAGI-1 with full history keeps 0.64 [0.45, 0.83] at 30 s ($+0.30$ [0.17, 0.43] over the released window, study 18) and [x] at 60 s. These shares are still falling at 60 s, slowly: from 30 to 60 s the re-rotated sink goes from 0.36 to 0.21 (study 20), Rolling Forcing from 0.59 to 0.45 and FramePack from 0.66 to 0.51. [Study 24, 180 s: whether they level off.] Keeping the opening does not stop the drift the openings share: MAGI-1's fresh rollouts brighten by 15–20 L^* under either setting, so full history preserves the difference between openings, not their level.

Sinks at their absolute positions: bursts. Without re-rotation a sink keeps more, and depth helps: no sink, one, three and six frames keep 0.05, 0.29, 0.61 and 0.76 at 60 s ($+0.15$ [0.11, 0.20] for six vs three; studies 7, 10), and LongLive's sink 0.45 against -0.01. Much of this is positional. Persistence jumps at the same times, near 17, 33 and 51 s, in every configuration with a sink at its absolute positions, across five Self Forcing checkpoints and LongLive (Appendix Fig. 5b, c), and never with a re-rotated sink or without one; the returns of the opening's scene coincide with these jumps (Section 4.1). Re-rotation lowers persistence at 60 s from 0.61 to 0.16 (-0.44 [-0.54, -0.35], preregistered) and, between the bursts (40–50 s), by -0.27 [-0.38, -0.16]: the absolute position helps even there. An attention probe accounts for the timing (exploratory; Fig. 4a): for the queries of a chunk we computed the share of attention the sink would receive if its keys sat at each query–sink distance from 3 to 300 latent frames (4 openings, 20 chunks each, all layers and denoising passes). The share peaks sharply at distances of 71, 134, 203 and 266 latent frames (0.11–0.15 against a median of 0.07), at the same distances in every chunk and opening, so the peaks belong to the model's positional code rather than to content. A sink at its original positions passes the first three peaks 16.5, 32.3 and 49.5 s after the opening, the times of the bursts. A second probe on four other openings, out to 723 frames (180 s), reproduces the curve ($r = 0.999$) and, by a peak rule fixed before its output was read, finds 13 further peaks after 60 s, preregistered as predicted burst times for the 180-s videos of study 24 (study 26). [Study 26: bursts at the 13 predicted onsets, contrast [x] (Self Forcing) and [x] (LongLive), [x] more than at control onsets between them.] LoL [21] relates a similar failure, "sink collapse", to rotary phases [51, 52]. [Study 25: moving the sink 18 or 36 frames further back moves the bursts by [x] s; held at a peak it keeps [x], at a trough [x].]

An action-conditioned world model. Matrix-Game-2.0 [50], driven by keyboard and mouse actions with no text, fits the same picture (study 15): a shift held only in its 6-frame context fades (-0.24 [-0.35, -0.14] from 5 to 30 s, preregistered; 7 scenes, as the model often collapses to black), while a shift sent only through

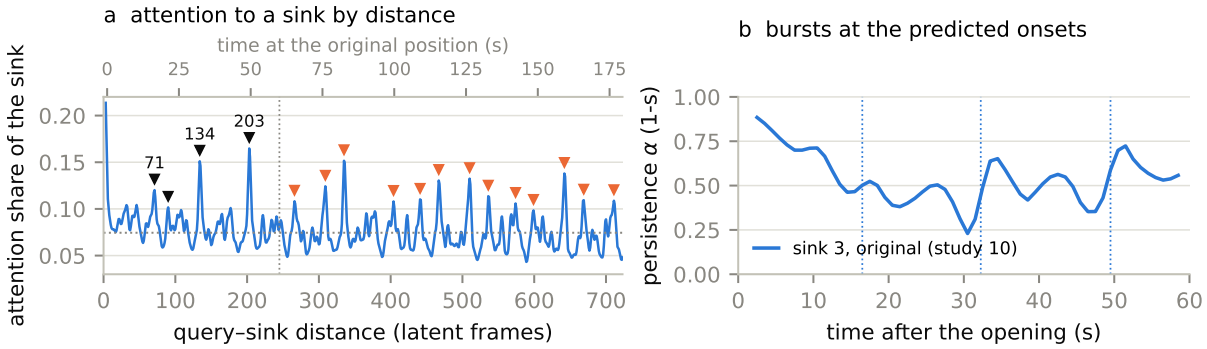


Figure 4: Bursts happen where the sink draws attention. (a) Share of attention that Self Forcing's queries would give to a three-frame sink at each query–sink distance (exploratory probe; 4 openings, 6 chunks each, all layers and passes; a first probe on 4 other openings to distance 300 gives the same curve). Triangles: major peaks (prominence ≥ 0.04); orange: the 13 after 60 s, preregistered as burst predictions (study 26). Top axis: when a sink at its original positions is at that distance. (b) Persistence of an L^* shift in 1-s windows (3-s moving average) with the sink at its original positions and moved 18 or 36 frames further back (study 25); dotted lines: burst onsets predicted from the peaks in (a).

its CLIP image embedding, which conditions every chunk, persists (0.43 [0.13, 0.71] at 30 s, Holm $p = 0.075$; 0.25 [0.17, 0.36] at 15 s over 20 scenes, exploratory; Appendix Fig. 7d).

A phenomenological recursion. A linear recursion of a chunk's mean statistic on the window, the sink and the prompt's default describes the first 20 s of these curves. Its prompt weight, $w_p \approx 0.08$ from a fit to 120-s curves, matched a bit-exact cache-replay measurement that swaps only the text conditioning for one chunk (0.088 [0.045, 0.149] at 15 s, study 14). It does not hold beyond 20 s and explains neither the share of re-rotated sinks nor the bursts (Appendix D).

6 Two levers

Equation 1 suggests two ways to keep a long rollout near its opening without retraining: raise α, or move μ toward the opening.

Raising α: a content-aware correction. At denoising step 1 of 4, the noisy chunk is denoised once more with a temporary cache that holds only the anchor (the opening chunk). The difference between the per-channel spatial means of the two predictions (the DC tone) is added to the real prediction, scaled by β; the main cache and the random stream are untouched. In a preregistered confirmation on 40 held-out prompts × 4 seeds (study 2), colour drift D fell by 1.48 [1.19, 1.76] at $\beta = 1$ and by 1.78 [1.45, 2.11] at $\beta = 1.5$, with structure drift K non-inferior, and in the relaxation study the correction raised Self Forcing's persistence from 0.28 to 0.40 (+0.12 [0.05, 0.20], preregistered). Table 1 compares it with two baselines. Post-hoc colour matching to the anchor reaches a similar D but, by construction, removes legitimate global changes too: on 7 held-out prompts with real brightness transitions (tunnels, garages, clouds), matching keeps 7–12% of the transition and the correction a median of 83% (per-prompt values from 0.58 to 1.71, so on some prompts it amplifies the change). A deeper attention sink, as in Deep Forcing [9] (study 13, bit-identical pairing), is a strong alternative: nine sink frames cut D by more than the correction at $\beta = 1$ (−0.45 [-0.75, -0.15]) with smaller boundary jumps, but a sink pulls everything toward the opening, and in two of the seven transition prompts it erased most of a legitimate change (7% and 19% kept, against 151% and 147% for the correction). The correction costs one extra forward pass per chunk (about 25% more compute) and more frame-to-frame jitter; it is the more conservative choice when real changes matter. Correcting at step 0 removes as much drift but erases tunnels, and steps 2–3 do nothing, consistent with global tone being fixed at the earliest denoising steps [41, 42].

Table 1: Correction versus post-hoc colour matching and deeper sinks (study 2 confirmation set: 40 held-out prompts $\times$ 4 seeds; transitions: 7 held-out prompts with legitimate brightness changes). Transitions kept: median over prompts of the corrected over the uncorrected first-to-last brightness change, prompt-bootstrap 95% CI.

Method	ΔD [95% CI]	transitions kept	jitter	boundary jumps
Content-aware correction, $\beta = 1$	-1.48 [-1.76, -1.19]	0.83 [0.67, 1.47]	$\times 1.13$	$\times 1.32$
Content-aware correction, $\beta = 1.5$	-1.78 [-2.11, -1.45]	0.93 [0.83, 1.48]	$\times 1.67$	$\times 2.40$
Post-hoc mean matching	-0.89 [-1.29, -0.50]	0.07	$\times 0.15$	$\times 0.12$
Post-hoc Reinhard [20]	-1.58 [-1.96, -1.19]	0.12	$\times 0.13$	$\times 0.10$
Deeper sink, 6 frames (study 13)	-1.30 [-1.68, -0.93]	0.70 [0.58, 1.12]	$\times 1.04$	$\times 0.98$
Deeper sink, 9 frames (study 13)	-1.93 [-2.31, -1.55]	0.59 [0.19, 1.09]	$\times 1.16$	$\times 1.09$

Moving μ: prompts. Since the prompt sets the default, lighting words can be chosen so that the default matches the opening. Choosing the lighting sentence whose *measured* default is nearest the opening's brightness reduces $|\Delta L|$ at 31 s by 26% on Self Forcing (-2.45 [-4.08, -1.04], Holm $p = 0.013$, study 4). A free-form Qwen2.5-VL [18] caption of the opening does not reduce brightness drift (study 3; it calls half of the normally lit frames "dim"), and the other five preregistered prompt tests failed, including LongLive with its own calibration (study 5). The mapping from words to brightness is model-specific and must be measured per model; Appendix F shows how well Eq. 1 forecasts these outcomes.

7 Discussion

Any autoregressive sampler with finite memory that is ergodic under a fixed conditioning will eventually forget its initial state, so that video rollouts relax is expected; what we measure is how much is kept and what decides it. Across nine models the answer is whether the opening stays in the context. Once it leaves a key–value window it is forgotten, and overlapping frames only delay this; an opening that stays (a re-rotated sink, an anchor, a per-section prefix, full history) keeps a share whose size depends on the model [and its 180-s behaviour]. The larger shares and returning scenes of sinks at their absolute positions are positional bursts [at predictable distances]. The default is set by the conditioning, for tone and for scene content, and models differ in which part of the conditioning carries which attribute: FramePack's image embedding sets the scene but not its brightness. Fresh rollouts of the two models trained on their own long rollouts barely drift, though they still forget (exploratory).

A high α is not always better. Under a conflicting prompt (a snowy opening, a city prompt), forgetting the opening is following the prompt, and an anchor that keeps the opening also passes its errors forward (SkyReels amplifies the shift in 5 of 12 scenes). Persistence is better treated as a property to choose and to report at a stated horizon, together with the default and its offset; our protocol measures both cheaply and would make long-video models comparable where a single quality score does not. For world models, a model conditioned on a real frame and a description drifts toward the description's default unless it re-conditions on the frame. In Self Forcing the mechanism that keeps the opening's scene is the unstable, positional one: keeping the opening's content while forgetting its errors is the open problem.

Limitations. Six of the nine models share the Wan2.1-1.3B architecture and a seventh is a student of its 14B sibling; FramePack and MAGI-1 have other bases. Content relaxation was replicated on FramePack, but the prompt dependence, the recursion and the decomposition rest on Wan-based models, and the world model collapses after 20–30 s. Interventions shift global tone or select a scene class (CLIP scores are a coarse proxy for content); openings are held images; scene content is compared across openings rather than shifted within one; most configurations use one seed per scene (the Self Forcing sink results on tone and content replicate with three more); Holm correction is applied within studies only; the correction was not compared with Pathwise test-time correction or FreqForcing.

References

[1] T. Yin, Q. Zhang, R. Zhang, W. T. Freeman, F. Durand, E. Shechtman, X. Huang. From slow bidirectional to fast autoregressive video diffusion models. CVPR 2025 (arXiv:2412.07772).

[2] X. Huang, Z. Li, G. He, M. Zhou, E. Shechtman. Self Forcing: Bridging the train-test gap in autoregressive video diffusion. arXiv:2506.08009, 2025.

[3] B. Chen, D. Martí Monsó, Y. Du, M. Simchowitz, R. Tedrake, V. Sitzmann. Diffusion Forcing: Next-token prediction meets full-sequence diffusion. NeurIPS 2024 (arXiv:2407.01392).

[4] Wan Team. Wan: Open and advanced large-scale video generative models. arXiv:2503.20314, 2025.

[5] T. Yin, M. Gharbi, R. Zhang, E. Shechtman, F. Durand, W. T. Freeman, T. Park. One-step diffusion with distribution matching distillation. CVPR 2024 (arXiv:2311.18828).

[6] M. Zhou, H. Zheng, Z. Wang, M. Yin, H. Huang. Score identity distillation: Exponentially fast distillation of pretrained diffusion models for one-step generation. ICML 2024 (arXiv:2404.04057).

[7] S. Yang, W. Huang, R. Chu, Y. Xiao, Y. Zhao, X. Wang, M. Li, E. Xie, Y. Chen, Y. Lu, S. Han, Y. Chen. LongLive: Real-time interactive long video generation. arXiv:2509.22622, 2025.

[8] K. Liu, W. Hu, J. Xu, Y. Shan, S. Lu. Rolling Forcing: Autoregressive long video diffusion in real time. arXiv:2509.25161, 2025.

[9] J. Yi, W. Jang, P. H. Cho, J. Nam, H. Yoon, S. Kim. Deep Forcing: Training-free long video generation with deep sink and participative compression. arXiv:2512.05081, 2025.

[10] SkyReels Team. SkyReels-V2: Infinite-length film generative model. arXiv:2504.13074, 2025.

[11] G. Xiao, Y. Tian, B. Chen, S. Han, M. Lewis. Efficient streaming language models with attention sinks. ICLR 2024 (arXiv:2309.17453).

[12] X. Xiang, Z. Duan, G. Zhang, H. Zhang, Z. Gao, J. Wu, S. Zhang, T. Wang, Q. Fan, C. Guo. Pathwise test-time correction for autoregressive long video generation. arXiv:2602.05871, 2026.

[13] J. Li, L. Liang, L. Kong, Y. Zhang. FreqForcing: Autoregressive long video generation via spectral self-anchoring. arXiv:2607.27110, 2026.

[14] J. Wang, F. Zhang, X. Li, V. Y. F. Tan, T. Pang, C. Du, A. Sun, Z. Yang. Error analyses of auto-regressive video diffusion models: A unified framework. arXiv:2503.10704, 2025.

[15] Z. Huang et al. VBench: Comprehensive benchmark suite for video generative models. CVPR 2024 (arXiv:2311.17982).

[16] J. Ke, Q. Wang, Y. Wang, P. Milanfar, F. Yang. MUSIQ: Multi-scale image quality transformer. ICCV 2021.

[17] M. Oquab et al. DINOv2: Learning robust visual features without supervision. TMLR 2024 (arXiv:2304.07193).

[18] Qwen Team. Qwen2.5-VL technical report. arXiv:2502.13923, 2025.

[19] D. Dauner et al. NAVSIM: Data-driven non-reactive autonomous vehicle simulation and benchmarking. NeurIPS 2024.

[20] E. Reinhard, M. Ashikhmin, B. Gooch, P. Shirley. Color transfer between images. IEEE CG&A, 2001.

[21] J. Cui, J. Wu, M. Li, T. Yang, X. Li, R. Wang, A. Bai, Y. Ban, C.-J. Hsieh. LoL: Longer than longer, scaling video generation to hour. arXiv:2601.16914, 2026.

[22] J. Cui, J. Wu, M. Li, T. Yang, X. Li, R. Wang, A. Bai, Y. Ban, C.-J. Hsieh. Self-Forcing++: Towards minute-scale high-quality video generation. arXiv:2510.02283, 2025.

[23] L. Zhang, S. Cai, M. Li, G. Wetzstein, M. Agrawala. Frame context packing and drift prevention in next-frame-prediction video diffusion models. arXiv:2504.12626, 2025.

[24] W. Kong et al. HunyuanVideo: A systematic framework for large video generative models. arXiv:2412.03603, 2024.

[25] J. Kim, J. Kang, J. Choi, B. Han. FIFO-Diffusion: Generating infinite videos from text without training. NeurIPS 2024.

[26] R. Henschel et al. StreamingT2V: Consistent, dynamic, and extendable long video generation from text. CVPR 2025 (arXiv:2403.14773).

[27] H. Qiu, M. Xia, Y. Zhang, Y. He, X. Wang, Y. Shan, Z. Liu. FreeNoise: Tuning-free longer video diffusion via noise rescheduling. ICLR 2024.

[28] Y. Lu, Y. Liang, L. Zhu, Y. Yang. FreeLong: Training-free long video generation with SpectralBlend temporal attention. NeurIPS 2024.

[29] D. Ruhe, J. Heek, T. Salimans, E. Hoogeboom. Rolling diffusion models. ICML 2024.

[30] Y. Jin et al. Pyramidal flow matching for efficient video generative modeling. ICLR 2025.

[31] Sand AI. MAGI-1: Autoregressive video generation at scale. arXiv:2505.13211, 2025.

[32] Krea AI. Krea Realtime 14B: real-time, long-form AI video generation. Technical blog post and model release, https://www.krea.ai/blog/krea-realtime-14b, 2025.

[33] K. Song, B. Chen, M. Simchowitz, Y. Du, R. Tedrake, V. Sitzmann. History-guided video diffusion. ICML 2025 (arXiv:2502.06764).

[34] S. Bengio, O. Vinyals, N. Jaitly, N. Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. NeurIPS 2015.

[35] S. Ross, G. Gordon, D. Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. AISTATS 2011.

[36] M. Ning, E. Sangineto, A. Porrello, S. Calderara, R. Cucchiara. Input perturbation reduces exposure bias in diffusion models. ICML 2023.

[37] M. Ning, M. Li, J. Su, A. A. Salah, I. O. Ertugrul. Elucidating the exposure bias in diffusion models. ICLR 2024.

[38] D. Valevski, Y. Leviathan, M. Arar, S. Fruchter. Diffusion models are real-time game engines. ICLR 2025.

[39] E. Alonso, A. Jelley, V. Micheli, A. Kanervisto, A. Storkey, T. Pearce, F. Fleuret. Diffusion for world modeling: Visual details matter in Atari. NeurIPS 2024.

[40] S. Gao, J. Yang, L. Chen, K. Chitta, Y. Qiu, A. Geiger, J. Zhang, H. Li. Vista: A generalizable driving world model with high fidelity and versatile controllability. NeurIPS 2024.

[41] S. Lin, B. Liu, J. Li, X. Yang. Common diffusion noise schedules and sample steps are flawed. WACV 2024.

[42] N. Guttenberg. Diffusion with offset noise. CrossLabs blog, 2023.

[43] X. Gu, T. Pang, C. Du, Q. Liu, F. Zhang, C. Du, Y. Wang, M. Lin. When attention sink emerges in language models: An empirical view. ICLR 2025.

[44] T. Darcet, M. Oquab, J. Mairal, P. Bojanowski. Vision transformers need registers. ICLR 2024.

[45] K. Li, T. Liu, N. Bashkansky, D. Bau, F. Viégas, H. Pfister, M. Wattenberg. Measuring and controlling instruction (in)stability in language model dialogs. COLM 2024.

[46] P. D. Oldham. A note on the analysis of repeated measurements of the same subjects. Journal of Chronic Diseases 15:969–977, 1962.

[47] A. G. Barnett, J. C. van der Pols, A. J. Dobson. Regression to the mean: what it is and how to deal with it. International Journal of Epidemiology 34(1):215–220, 2005.

[48] A. Radford et al. Learning transferable visual models from natural language supervision. ICML 2021.

[49] X. Zhai, B. Mustafa, A. Kolesnikov, L. Beyer. Sigmoid loss for language image pre-training. ICCV 2023.

[50] X. He, C. Peng, Z. Liu, B. Wang et al. Matrix-Game 2.0: An open-source, real-time, and streaming interactive world model. arXiv:2508.13009, 2025.

[51] M. Zhao, G. He, Y. Chen, H. Zhu, C. Li, J. Zhu. RIFLEx: A free lunch for length extrapolation in video diffusion transformers. ICML 2025 (arXiv:2502.15894).

[52] F. Barbero, A. Vitvitskyi, C. Perivolaropoulos, R. Pascanu, P. Veličković. Round and round we go! What makes rotary positional encodings useful? arXiv:2410.06205, 2024.

A Models, preregistered tests, deviations and withdrawn claims

Table 2: Configurations. Wan-based cache models generate 832×480 at 16 fps in chunks of 3 latent frames (0.75 s); window and sink sizes are in latent frames.

Configuration	Base; training	Steps	Context of the opening
Self Forcing DMD, SiD, SiD v2, GAN [2, 5, 6]	Wan2.1-1.3B; self-rollout on 5-s clips	4	window 21; sink 0/1/3/6, original or re-rotated positions
Self Forcing 10 s	Wan2.1-1.3B; self-rollout on 10-s clips	4	window 21, sink 3
CausVid [1]	Wan2.1-1.3B; teacher-forced DMD	3	window 21, sink 0/3
LongLive [7]	Wan2.1-1.3B; long-rollout tuning	4	window 12, sink 0/3
Rolling Forcing [8]	Wan2.1-1.3B; rolling joint denoising	5	anchor block + 21
Krea Realtime 14B [32]	Wan2.1-14B; Self Forcing distillation	4	key–value cache recomputed from the last 3 latent frames; optional first-frame anchor (1 of the 3)
SkyReels-V2-DF [10]	Wan2.1-1.3B architecture; diffusion forcing; 544p, 24 fps	30 + CFG	17 (or 5) re-denoised overlapping frames
Matrix-Game-2.0 [50]	Wan2.1-1.3B architecture, no text; actions; 352×640, 25 fps	3	first latent frame only, window 6, no sink; CLIP image embedding
FramePack-F1 / I2V [23]	HunyuanVideo 13B; 480p, 30 fps	25	clean prefix in every section; SigLIP image embedding
MAGI-1 4.5B-distill [31]	own architecture (chunk-wise autoregressive diffusion transformer); 432×768, 24 fps, chunks of 1 s	64, pipelined	released: window of 4 earlier chunks, no sink; or full history

Deviations from the preregistrations.

- *Study 1* was frozen before analysis but after generation of its main videos had started, and its arms were not pinned to one GPU architecture: 62% of its nominal seed pairs ran on different architectures. Its preregistered Wald intervals are reported; the prompt-cluster bootstrap interval for the main contrast is wider ([-10.45, -5.39] for D) but leads to the same conclusion.
- *Study 3:* a preempted job resumed on another GPU type and split four LongLive openings across architectures. We regenerated the affected arms on the original architecture; conclusions are unchanged.
- *Study 6:* the ODE-initialised Self Forcing student was replaced by SiD v2 before generation (it collapses to blur after the opening). GPU pinning was replaced by an architecture guard that regenerates a task's outputs when a resumed job lands on another architecture; paired arms of each preregistered contrast always share a task. Exploratory configurations (sink 1; correction on LongLive and CausVid) were added after freezing but before any data, from separate task lists, so their contrasts with the preregistered arms are not guaranteed to share an architecture. Prompt-only reference rollouts were renamed after generation; no analysis code changed.
- *Study 7* reuses the 24 scenes of study 6, including the 6 whose 120-s curves motivated it. Restricted to the 18 other scenes, the sink effect is +0.53 [0.42, 0.63] (Self Forcing) and +0.18 [0.13, 0.24] (CausVid).
- *Studies 10 and 11* were run on two non-preemptible 4-GPU jobs and A40-only array tasks instead of the preregistered all-queue arrays, because no large GPU was free on the shared queue; all their videos ran on A40 GPUs, so pairing is unaffected. The other studies ran as preregistered.

Withdrawn or revised claims. During the project we withdrew four claims after checks: that MUSIQ rewards collapsed videos (it is merely insensitive); that the sink stabilises the default (trend difference +0.5 [-2.4, 3.5]); that the model follows its context by 76% (true only for artificial shifts); and that early dynamics forecast the final brightness. The recursion's predicted Self Forcing plateau of 0.25–0.54 was missed (0.58 at 60 s in study 7).

B Drift metrics for prompt-only rollouts

For rollouts without a held opening, drift is measured against 10 (Self Forcing) or 5 (LongLive) fresh 5-s generations of the same prompt. D is the per-frame Mahalanobis distance of (Lab means, saturation, log sharpness) from the fresh frames' distribution, and K the mean cosine distance of a frame's DINOv2 feature to its 5 nearest fresh-frame features, normalised by the fresh clips' own value; both are averaged over seconds 29–31. Transitions kept is the median over held-out transition prompts of the corrected over the uncorrected first-to-last brightness change; values above 1 mean the correction amplified the change.

C Robustness

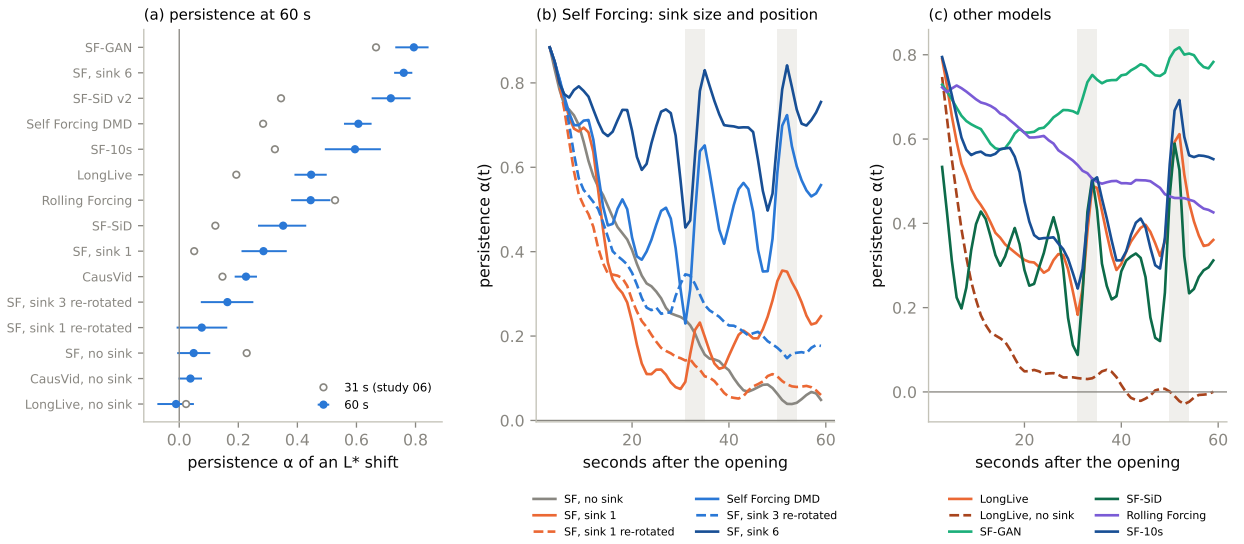


Figure 5: Persistence at 60 s (study 10; CausVid from study 7). (a) α of an L* shift with scene-bootstrap 95% CIs; hollow markers: 31 s (study 6). (b) Self Forcing with sinks of 0, 1, 3 and 6 latent frames, at their original positions or re-rotated to sit just before the window (dashed). (c) Other models. Curves: 1-s windows, 3-s moving average; shaded: 31–35 s and 50–54 s, where sinks of three or more frames at their original positions produce bursts. The 60-s endpoint window, [50, 60) s, overlaps the second burst; Appendix C reports a burst-free window.

Linearity. For 12 of 14 configurations the persistence estimated from ±8 and from ±16 shifts differs by at most 0.09, with intervals including 0. SiD and the sink-1 configuration give near-zero slopes for small shifts (−0.10, −0.15) and positive ones for large shifts (0.18, 0.10); LongLive with a sink keeps a darkening better than a brightening (0.36 vs 0.04). The linear law is a first-order description.

The 60-s endpoint and the positional bursts. The preregistered 60-s window, 50–60 s, overlaps the second burst. Over 40–50 s, Self Forcing keeps 0.48 with a three-frame sink and 0.20 with a re-rotated one (difference −0.27 [-0.38, -0.16]); over the full period 35–60 s, 0.53 and 0.19. Sink depth is monotone in every window (40–50 s: no sink 0.08, one frame 0.23, three 0.48, six 0.63). Some curves are not monotone decays

(GAN and SiD v2 lose part of the shift early and regain it), and Rolling Forcing is still declining at 60 s. In the 120-s subset of study 6 (6 scenes), Self Forcing with a sink keeps 0.54, 0.48 and 0.44 at 60, 90 and 118 s. No value we report is shown to be an asymptote; x_∞ in Eq. 1 denotes the long-run tendency, and α is always reported with its horizon.

Re-rotated sinks across studies. A re-rotated three-frame sink keeps 0.16 at 60 s in study 10 and 0.23 in study 16 because 9 of the 24 scenes ran on another GPU architecture in study 16, which amounts to resampling them; all 72 videos repeated on the same architecture were bit-identical.

The default and fresh rollouts. Across 10 configurations, the zero-drift level correlates $r = 0.93$ with where prompt-only rollouts end, but the slope is 1.58 and the mean absolute difference 4.9 L^* (largest for the GAN and SiD v2 checkpoints, whose zero-drift levels are extrapolated from slopes far from 0). Fresh rollouts give the direction and rough size of the default, not its exact value.

Persistence depends on the prompt. In the lighting experiment, the slope of the 31-s change on the opening differs between lighting prompts on Self Forcing (prompt × opening interaction, opening-clustered Wald test $p = 0.001$): persistence is 0.34 under sunny, 0.24 overcast, 0.17 dusk and 0.05 night prompts. On LongLive the interaction is weaker ($p = 0.053$). Dark defaults compress the range of brightness the model produces, which may explain part of this. Study 12 confirms the prompt dependence in four new scene types (Section 4.2).

Content scores do not depend on the encoder. Every content result uses CLIP ViT-L/14 class log-odds. We rescored all content videos with SigLIP so400m-patch14-384 [49], with its own text embeddings of the same seven labels, and reran each frozen analysis unchanged (Table 6). Every verdict agrees. Estimates in κ agree within 0.03; raw log-odds differ in scale between encoders. This includes FramePack's embedding route, whose conditioning encoder is itself a SigLIP model: CLIP and SigLIP give it the same content share (0.23 and 0.25). Class scores might also carry global colour (snow is bright, sand is yellow). On frames whose CIELAB mean and standard deviation are matched to the held opening's before encoding, the city-versus-class-prompt difference of study 9 is 9.0 $[7.9, 10.2]$ log-odds (9.3 without matching), and the four tests of study 21 give +0.45, −0.38, +0.08 and −0.32 (+0.48, −0.40, +0.08, −0.34 without).

Real and generated openings. On Self Forcing, real and generated openings fall on the same line ($p = 0.80$); on LongLive, generated openings end 4.3 L^* brighter than real ones of equal brightness ($p = 0.03$).

D The recursion and the replay

Model a chunk's mean statistic as $x_{k+1} = w_w\,\bar{x}_{\text{window}} + w_s\,x_{\text{sink}} + w_p\,\mu$ with $w_w + w_s + w_p = 1$, where the opening counts as a window chunk until it rolls out and μ is the prompt's default. Without a sink the opening is eventually forgotten; with one, $\alpha_\infty = w_s/(w_s + w_p)$, so a small sink weight keeps a large share if the prompt's pull is also small. Fitted to 120-s curves, the recursion gives $w_s = 0.060$ $[0.021, 0.097]$ and $w_w = 0.86$, implying $w_p \approx 0.08$, which we preregistered as the expected value of the replay measurement (the preregistered test was $w_p > 0$). Study 14 measured 0.088 $[0.045, 0.149]$ at 15 s. With a sink the prompt's pull rises to 0.26 $[0.17, 0.36]$ at 30 s (preregistered test of time invariance, +0.16 $[0.10, 0.24]$), so the recursion holds for about 20 s. Bit-exact replay of the key–value cache measures the recursion's weights directly, by regenerating one chunk from identical noise after changing only the sink, the window or the text conditioning: $w_s = 0.053$ $[0.040, 0.066]$ for synthetic tone shifts (0.036 in a dose–response design), and $w_w = 0.76$ (synthetic) to 0.90–0.99 (natural differences), in 10 development prompts × 2 seeds. The prompt's pull measured in study 14 is proportional to the difference between the two prompts' defaults, and a prompt with the same default has no effect (Fig. 6b, c). The three weights sum to 1.00 with the fitted w_w, but to 0.90 or 1.13 with the replay-measured ones, and α_∞ is 0.29–0.41 depending on which w_s is used. Without a sink the measured pull is smaller (0.035 at 15 s), so w_p is not a property of the prompt alone, and fitted weights depend on the horizon. Re-rotated sinks of three or six frames keep 0.16–0.23, below what the recursion implies; a sink at its absolute positions keeps 0.61, above it.

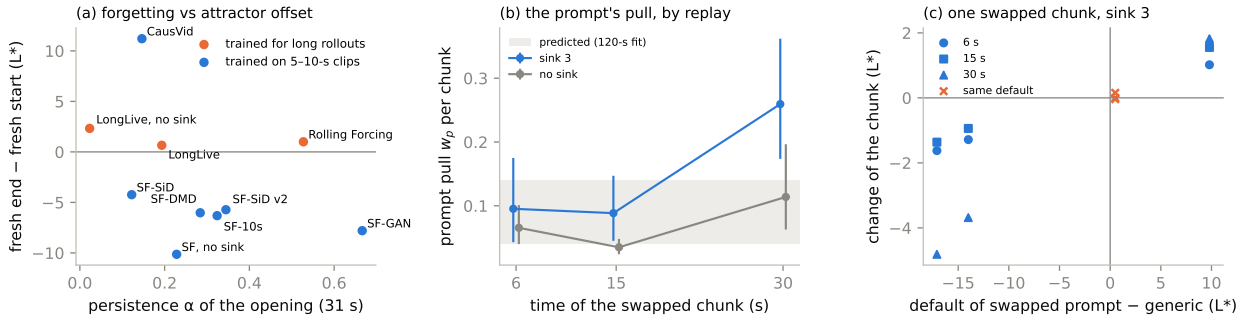


Figure 6: (a) Two components of drift: persistence of the opening (study 6) against the attractor offset of fresh rollouts (end minus start brightness, 4 seeds); models trained on their own long rollouts have almost no offset but still forget. (b) The prompt's per-chunk pull measured by bit-exact replay (study 14; 16 seeds; 95% CIs): one chunk regenerated from identical cache, noise and random state with only the text conditioning swapped; grey: the value implied by the recursion fitted to 120-s curves, preregistered for 15 s. (c) The swapped chunk's change against the difference of the prompts' defaults; a prompt with the same default has no effect.

E Models outside the Wan cache family

F The default and the forecast

How well does the law forecast? Taking α and the generic default from the relaxation study and the lighting defaults from the lighting experiment (different scenes and openings), $\Delta L \approx (1-\alpha)(\mu - L_{\text{open}})$ predicts the 31-s brightness change of each opening in the calibrated-lighting study with $r = 0.82$ (Self Forcing) and 0.77 (LongLive), mean absolute error 3.8 L*. Most of this correlation reflects the opening's brightness. The forecast of the *benefit* of calibration per opening is weaker ($r = 0.46$ and 0.58), and its mean is overpredicted (-4.5 predicted vs -3.2 and -2.8 observed L*). The law is a first-order guide to a prompt choice, not a precise forecast.

G Pitfalls for evaluating long-video models

1. **The GPU architecture changes the video.** With bf16, outputs are bit-identical within an architecture and differ across A40, A100/A800 and L40S/RTX 6000 Ada. Paired designs must pin or guard the architecture.
2. **Resumed jobs can silently break pairing.** Generators that skip existing outputs resume a preempted job wherever it lands.
3. **Image-quality scores miss drift.** MUSIQ, used by VBench for imaging quality, varies only between 66 and 69 across context strategies whose colour drift differs threefold.
4. **Natural variation is too small to estimate persistence.** Within a prompt, seeds start only about 2 L* apart, so within-prompt slopes of end on start brightness have intervals of about ±0.4; controlled shifts are needed.
5. **Change-on-baseline regressions mislead.** Regressing change on a baseline measured with error produces spurious regression to the mean [46, 47]. Held openings are measured almost without error, and the interventional design removes the issue.

H Context strategies (study 1)

A preregistered factorial study of 10 window and sink strategies on Self Forcing (40 prompts × 4 seeds, 31 s) found that a 3-frame sink cuts colour drift D by 7.56 $[6.81, 8.32]$ and DINOv2 drift K by 0.90; that

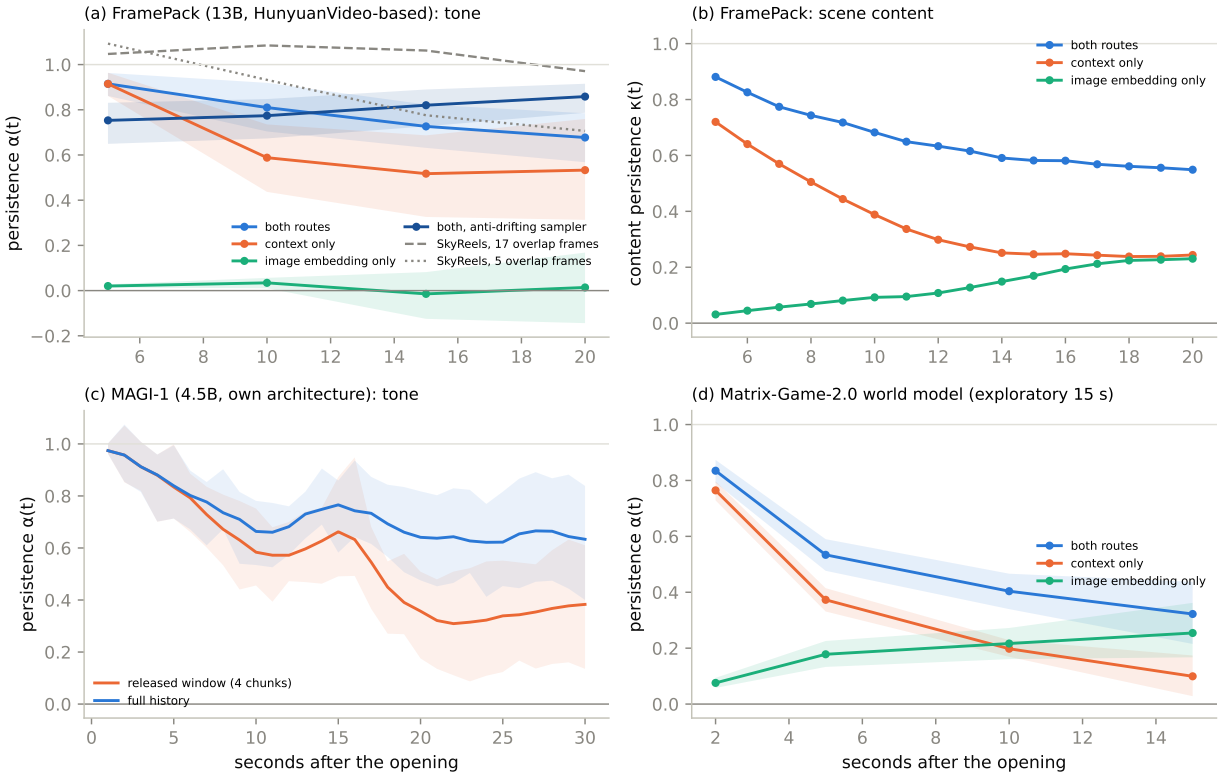


Figure 7: (a) FramePack (study 11, 12 scenes): the opening's brightness shift sent through both routes, only the clean context prefix, or only the SigLIP image embedding; the anti-drifting sampler; SkyReels-V2 with 17 and 5 overlapping frames for reference. (b) FramePack, scene content (study 17, 24 openings, city prompt): with a city frame in the context, the opening's embedding alone pulls the video toward the opening's class. (c) MAGI-1 (study 18, 12 scenes): the released window of four 1-s chunks against full history (1-s windows, 3-s moving average). (d) Matrix-Game-2.0, an action-conditioned world model without text (study 15, exploratory 15-s analysis, 20 scenes): the context latent leaves the window and is forgotten, while the rollout drifts toward the CLIP-embedded opening. Bands: scene-bootstrap 95% CIs.

window size matters only without a sink; and that re-rotating the sink's positions (which keeps the query–sink distance within the range seen in training) does not reduce D (-0.03 [-0.79, 0.72]) and increases K ($+0.36$). Darkening persists with a sink (last-2-s $z_L = -1.99$). D and K rise roughly linearly over 31 s under every strategy, without a visible plateau.

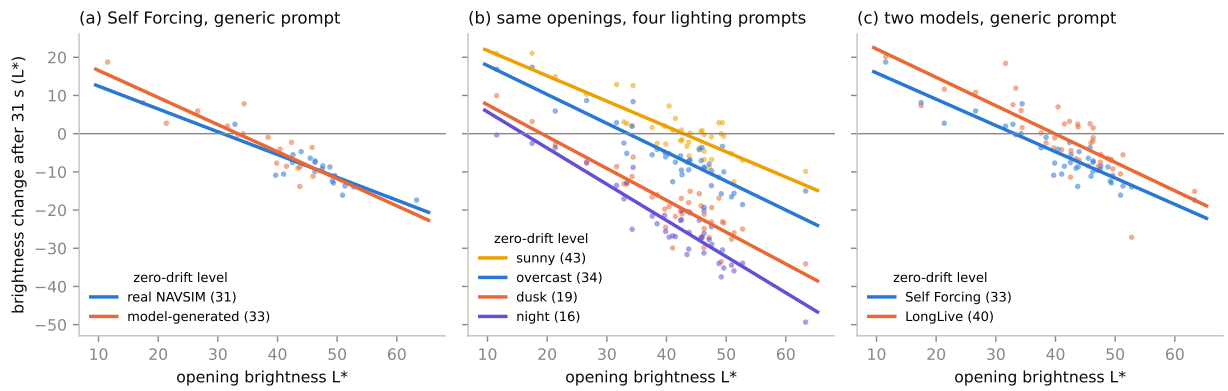


Figure 8: Brightness change after 31 s against the opening's brightness (one dot per opening, generic or lighting prompt): (a) real and generated openings follow one line; (b) four lighting prompts set four defaults (Self Forcing); (c) two models: similar slopes, different defaults.

Table 3: All preregistered primary tests and their outcomes. Intervals are Wald intervals from the preregistered mixed models in studies 1, 2 and 13 and bootstrap intervals otherwise; p is Holm-adjusted within each study (no correction across studies). Study 3 LongLive rows are as executed; after regenerating four cross-architecture openings the conclusions are unchanged; in study 1, 62% of nominal pairs crossed GPU architectures, which widens but does not bias its intervals. Units: D, K drift metrics; $|\Delta L|$ in L^*; α persistence, κ content persistence. Arms: G generic prompt, M VLM caption of the opening, C calibrated lighting plus caption, ttc content-aware correction. [‡]The t-test treats the default μ as fixed; the bootstrap interval also resamples the fresh starts that estimate it (pooled with seed 0: $[0.02, 0.15]$).

Study	Test	Model	Estimate [95% CI]	p_{Holm}	Outcome
1 Context	sink 3 vs none, D	SF	-7.56 [-8.32, -6.81]	< 0.001	supported
	sink 3 vs none, K	SF	-0.90 [-1.04, -0.76]	< 0.001	supported
	window × sink, D	SF	$+5.71$ [4.21, 7.22]	< 0.001	supported
	window × sink, K	SF	$+0.67$ [0.38, 0.95]	< 0.001	supported
2 Correction	ttc β=1, D	SF	-1.48 [-1.76, -1.19]	$< 10^{-23}$	supported
	ttc β=1.5, D	SF	-1.78 [-2.11, -1.45]	$< 10^{-24}$	supported
3 Caption	M − G, $\|\Delta L\|$	SF	-1.21 [-3.09, 0.36]	0.58	not supported
	M+ttc − M, $\|\Delta L\|$	SF	-1.00 [-1.89, -0.12]	0.14	not supported
	M − G, $\|\Delta L\|$	LL	-0.19 [-2.00, 1.63]	1.0	not supported
	M+ttc − M, $\|\Delta L\|$	LL	-0.36 [-1.48, 0.75]	1.0	not supported
4 Calibrated	C − M, $\|\Delta L\|$	SF	-1.24 [-2.54, 0.12]	0.25	not supported
	C − G, $\|\Delta L\|$	SF	-2.45 [-4.08, -1.04]	0.013	supported
	C − M, $\|\Delta L\|$	LL	-1.15 [-2.59, 0.29]	0.26	not supported
	C − G, $\|\Delta L\|$	LL	-1.34 [-3.19, 0.55]	0.26	not supported
5 Calibrated (LL scale)	C − M, $\|\Delta L\|$	LL	-1.36 [-3.06, 0.32]	0.26	not supported
	C − G, $\|\Delta L\|$	LL	-1.55 [-3.50, 0.39]	0.26	not supported
6 Relaxation	no sink − sink, α_L (31 s)	SF	-0.06 [-0.17, 0.05]	0.34	not supported
	no sink − sink, α_L (31 s)	LL	-0.17 [-0.28, -0.06]	0.013	supported
	ttc − none, α_L (31 s)	SF	$+0.12$ [0.05, 0.20]	0.013	supported
	no sink − sink, α_b (31 s)	SF	-0.26 [-0.34, -0.18]	< 0.001	supported
7 Sink at 60 s	sink − no sink, α_L	SF	$+0.55$ [0.45, 0.65]	$< 10^{-9}$	supported
	sink − no sink, α_L	CausVid	$+0.19$ [0.14, 0.24]	$< 10^{-6}$	supported
8 Overlap	5 − 17 frames, α_L (20 s)	SkyReels	-0.24 [-0.40, -0.09]	0.014	supported
9 Content	class − city prompt, class score (30 s)	SF	$+9.33$ [8.11, 10.51]	$< 10^{-9}$	supported
	sink − no sink, κ (30 s)	LL	-0.02 [-0.10, 0.07]	0.72	not supported
	sink − no sink, κ (60 s)	SF	$+0.48$ [0.39, 0.57]	$< 10^{-7}$	supported
10 Atlas	sink 6 − sink 3, α_L (60 s)	SF	$+0.15$ [0.11, 0.20]	$< 10^{-6}$	supported
	sink 1 − no sink, α_L (60 s)	SF	$+0.24$ [0.14, 0.34]	< 0.001	supported
	sink − no sink, α_L (60 s)	LL	$+0.46$ [0.39, 0.53]	$< 10^{-10}$	supported
	re-rotated − original sink, α_L (60 s)	SF	-0.44 [-0.54, -0.35]	$< 10^{-7}$	supported
11 FramePack	α (both routes, 20 s) > 0.5	FramePack-F1	0.68 [0.57, 0.79]	0.033	supported
	α (embedding only, 20 s) > 0	FramePack-F1	0.01 [-0.14, 0.17]	0.87	not supported
	both − context only, α (20 s)	FramePack-F1	$+0.14$ [-0.05, 0.33]	0.38	not supported
12 Prompts	sink − no sink, α_L (60 s), 4 domains	SF	$+0.53$ [0.45, 0.61]	$< 10^{-10}$	supported
	sink − no sink, α_L (30 s), 4 domains	LL	$+0.19$ [0.11, 0.28]	0.005	supported
	α_L differs between domains (ANOVA)	SF	$F = 3.12$	0.049	supported
13 Deep sink	sink 6, D	SF	-1.30 [-1.68, -0.93]	$< 10^{-6}$	supported
	sink 9, D	SF	-1.93 [-2.31, -1.55]	$< 10^{-6}$	supported
14 Prompt pull	$w_p > 0$ at 15 s (replay)	SF	$+0.088$ [0.045, 0.149]	0.006	supported
15 World model	ctx-only shift fades (30 vs 5 s)	Matrix-Game	-0.24 [-0.35, -0.14]	0.024	supported
	emb-only shift persists, α (30 s)	Matrix-Game	$+0.43$ [0.13, 0.71]	0.075	not supported
	both − ctx-only, α (30 s)	Matrix-Game	$+0.33$ [-0.07, 0.71]	0.18	not supported
16 Re-rotation	re-rotated − original, burst contrast R	SF	-0.29 [-0.37, -0.22]	$< 10^{-5}$	supported
	re-rotated − original sink, κ (60 s)	SF	-0.31 [-0.37, -0.25]	$< 10^{-6}$	supported
	re-rotated sink − none, κ (60 s)	SF	$+0.10$ [-0.00, 0.21]	0.11	not supported
	re-rotated − original sink 6, α_L (60 s)	SF	-0.54 [-0.62, -0.45]	$< 10^{-9}$	supported
	re-rotated sink 6 − sink 3, α_L (60 s)	SF	-0.00 [-0.11, 0.10]	0.94	not supported
	α (both routes, 60 s) > 0	FramePack-F1	0.51 [0.29, 0.71]	0.004	supported
	α (context only, 60 s) > 0	FramePack-F1	0.46 [0.28, 0.65]	0.003	supported
	5 frames: α (60 s) − α (20 s)	SkyReels	-0.57 [-0.79, -0.39]	0.001	supported
	17 frames: α (60 s) − α (20 s)	SkyReels	-0.72 [-0.81, -0.63]	$< 10^{-6}$	supported
17 FP content	content falls, class score (20 s)	FramePack-F1	-6.72 [-8.29, -5.20]	$< 10^{-5}$	supported
	κ (context only, 20 s) > 0	FramePack-F1	0.24 [0.12, 0.37]	0.001	supported
	κ (embedding only, 20 s) > 0	FramePack-F1	0.23 [0.16, 0.30]	$< 10^{-6}$	supported
	both − context only, κ (20 s)	FramePack-F1	$+0.31$ [0.20, 0.41]	< 0.001	supported
18 MAGI-1	window: α (30 s) − α (5 s)	MAGI-1	-0.52 [-0.70, -0.35]	< 0.001	supported
	full history − window, α (30 s)	MAGI-1	$+0.30$ [0.17, 0.43]	0.002	supported
	α (full history, 30 s) > 0	MAGI-1	0.64 [0.45, 0.83]	< 0.001	supported
20 Seeds	sink 3 − none, α (60 s)	SF	$+0.62$ [0.56, 0.67]	$< 10^{-15}$	supported
	re-rotated − original sink, α (60 s)	SF	-0.39 [-0.46, -0.32]	$< 10^{-8}$	supported
	re-rotated sink − none, α (60 s)	SF	$+0.22$ [0.15, 0.29]	$< 10^{-5}$	supported
21 Content seeds	sink 3 − none, κ (60 s)	SF	$+0.48$ [0.40, 0.56]	$< 10^{-10}$	supported
	re-rotated − original sink, κ (60 s)	SF	-0.40 [-0.47, -0.34]	$< 10^{-12}$	supported
	re-rotated sink − none, κ (60 s)	SF	$+0.08$ [-0.01, 0.16][‡]	0.006	supported
	re-rotated − original, burst contrast R	SF	-0.34 [-0.38, -0.28]	$< 10^{-10}$	supported
22 Krea	window: α (60 s) − α (10 s)	Krea 14B	-0.38 [-0.45, -0.31]	$< 10^{-9}$	supported
	anchor − window, α (60 s)	Krea 14B	$+0.08$ [-0.03, 0.18]	0.32	not supported
	α (anchor, 60 s) > 0	Krea 14B	0.03 [-0.03, 0.08]	0.37	not supported
	content falls, class score (55–60 s)	Krea 14B	-12.02 [-12.64, -11.37]	$< 10^{-15}$	supported
	anchor − window, κ (55–60 s)	Krea 14B	$+0.33$ [0.18, 0.47]	$< 10^{-5}$	supported

Table 4: Persistence over the same window, 15–20 s after the opening, for every model (exploratory; scene-bootstrap 95% CIs; Matrix-Game: scenes without a black second before 20 s).

Configuration	α	95% CI	scenes
SkyReels-V2-DF, 17 overlap frames	1.01	[0.89, 1.15]	12
FramePack-I2V, anti-drifting sampler	0.86	[0.79, 0.92]	12
SkyReels-V2-DF, 5 overlap frames	0.77	[0.64, 0.90]	12
Self Forcing DMD, sink 6	0.72	[0.65, 0.78]	24
FramePack-F1, both routes	0.68	[0.56, 0.78]	12
Rolling Forcing	0.64	[0.60, 0.67]	24
Self Forcing GAN	0.61	[0.54, 0.67]	24
Self Forcing 10-s	0.54	[0.46, 0.63]	24
FramePack-F1, context prefix only	0.53	[0.31, 0.76]	12
Self Forcing DMD, sink 3	0.49	[0.44, 0.53]	24
Self Forcing DMD, no sink	0.44	[0.38, 0.51]	24
Self Forcing DMD, sink 3 re-rotated	0.38	[0.28, 0.46]	24
Matrix-Game-2.0, both routes	0.37	[0.22, 0.52]	17
Self Forcing SiD	0.34	[0.24, 0.46]	24
LongLive, sink 3	0.33	[0.25, 0.40]	24
Self Forcing DMD, sink 1 re-rotated	0.33	[0.25, 0.41]	24
Self Forcing DMD, sink 1	0.29	[0.17, 0.41]	24
Self Forcing SiD v2	0.24	[0.11, 0.36]	24
Matrix-Game-2.0, embedding only	0.22	[0.06, 0.36]	17
CausVid, sink 3	0.21	[0.16, 0.26]	24
Matrix-Game-2.0, context only	0.10	[-0.02, 0.22]	17
LongLive, no sink	0.08	[0.04, 0.12]	24
CausVid, no sink	0.08	[0.04, 0.12]	24
FramePack-F1, embedding only	0.01	[-0.14, 0.17]	12

Table 5: Every global statistic relaxes toward a default (observational; 40 held openings, generic prompt, 31 s). Persistence = 1 + slope of the change on the opening value; r = correlation of the change with the opening value.

	Self Forcing		LongLive	
Statistic	persistence	r	persistence	r
L^*	0.31	−0.91	0.26	−0.81
a^*	0.38	−0.92	0.39	−0.90
b^*	0.22	−0.88	0.42	−0.80
saturation	0.54	−0.92	0.23	−0.95
log sharpness	0.37	−0.91	0.23	−0.94
L^* contrast	0.34	−0.82	0.37	−0.79

Table 6: Content results do not depend on the scoring encoder (exploratory). Each preregistered content test rerun with its frozen analysis on SigLIP so400m embeddings instead of CLIP ViT-L/14. Estimates in each encoder's log-odds or κ units; Holm p within each study; † not supported. The verdict agrees for 14 of 14 tests.

		CLIP		SigLIP	
Study	Test	est.	p	est.	p
09 H1	prompt sets content: $s_{\text{class}} - s_{\text{generic}}$ (30 s)	+9.33	<0.001	+13.24	<0.001
09 H2	LongLive sink keeps content: $\Delta\kappa$ (30 s)	-0.02	0.718†	+0.00	0.920†
09 H3	Self Forcing sink keeps content: $\Delta\kappa$ (60 s)	+0.48	<0.001	+0.48	<0.001
16 A1	late return is positional: ΔR	-0.29	<0.001	-0.28	<0.001
16 A2	re-rotation lowers content: $\Delta\kappa$ (60 s)	-0.31	<0.001	-0.32	<0.001
16 A3	re-rotated sink keeps content: $\Delta\kappa$ (60 s)	+0.10	0.113†	+0.09	0.099†
17 H1	FramePack content relaxes: $s - s_{\text{open}}$	-6.72	<0.001	-10.10	<0.001
17 H2	context prefix carries content: κ	+0.24	0.001	+0.22	<0.001
17 H3	SigLIP embedding carries content: κ	+0.23	<0.001	+0.25	<0.001
17 H4	embedding adds content: $\Delta\kappa$	+0.31	<0.001	+0.30	<0.001
21 H1	sink keeps content, 3 new seeds: $\Delta\kappa$ (60 s)	+0.48	<0.001	+0.50	<0.001
21 H2	re-rotation lowers content: $\Delta\kappa$ (60 s)	-0.40	<0.001	-0.40	<0.001
21 H3	re-rotated sink keeps content: $\Delta\kappa$ (60 s)	+0.08	0.006	+0.10	<0.001
21 H4	late return is positional: ΔR	-0.34	<0.001	-0.32	<0.001